

```
from tensorflow.keras.datasets import imdb
```

```
%pip install tensorflow
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.20.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (18.1.1)
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Requirement already satisfied: protobuf>=5.28.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (5.29.6)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow) (75.2.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.1.2)
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Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.13.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.0.2)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.16.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.5.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from astunparse>=1.6.0->tensorflow) (13.9.4)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow) (13.9.4)
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.10.1)
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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (2025.11.12)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.20.0->tensorflow) (3.7.0)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.20.0->tensorflow) (11.0.0)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.20.0->tensorflow) (0.7.0)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.20.0->tensorflow) (3.1.3)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from werkzeug>=1.0.1->tensorboard~2.20.0->tensorflow) (3.0.2)
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Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.2)
```

```
(train_data, train_label), (test_data, test_label) = imdb.load_data(num_words = 10000)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz
17464789/17464789 ————— 2s 0us/step
```

```
import numpy as np
```

```
def vectorize_sequences(sequences, dimensions = 10000):
    results = np.zeros((len(sequences), dimensions))
    for i, sequences in enumerate(sequences):
        results[i, sequences] = 1
    return results
```

```
x_train = vectorize_sequences(train_data)
y_train = vectorize_sequences(test_data)
```

```
y_train = np.asarray(train_label).astype('float32')
y_test = np.asarray(test_label).astype('float32')
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

```
model = Sequential()
model.add(Dense(16, input_shape=(10000, ), activation = "relu"))
model.add(Dense(16, activation = "relu"))
model.add(Dense(1, activation = "sigmoid"))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input`
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
model.compile(optimizer='adam', loss = 'mse', metrics = ['accuracy'])
```

```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 16)	160,016
dense_1 (Dense)	(None, 16)	272
dense_2 (Dense)	(None, 1)	17

Total params: 160,305 (626.19 KB)
Trainable params: 160,305 (626.19 KB)
Non-trainable params: 0 (0.00 B)

history = model.fit(x_train, y_train, validation_split = 0.3, epochs = 20, verbose = 1, batch_size = 512)

Epoch 1/20
35/35 ————— 7s 172ms/step - accuracy: 0.7837 - loss: 0.1814 - val_accuracy: 0.8675 - val_loss: 0.1155
Epoch 2/20
35/35 ————— 3s 85ms/step - accuracy: 0.9073 - loss: 0.0832 - val_accuracy: 0.8895 - val_loss: 0.0860
Epoch 3/20
35/35 ————— 4s 54ms/step - accuracy: 0.9388 - loss: 0.0556 - val_accuracy: 0.8944 - val_loss: 0.0811
Epoch 4/20
35/35 ————— 2s 44ms/step - accuracy: 0.9565 - loss: 0.0414 - val_accuracy: 0.8904 - val_loss: 0.0822
Epoch 5/20
35/35 ————— 3s 66ms/step - accuracy: 0.9693 - loss: 0.0318 - val_accuracy: 0.8892 - val_loss: 0.0829
Epoch 6/20
35/35 ————— 2s 46ms/step - accuracy: 0.9782 - loss: 0.0248 - val_accuracy: 0.8833 - val_loss: 0.0864
Epoch 7/20
35/35 ————— 1s 39ms/step - accuracy: 0.9849 - loss: 0.0192 - val_accuracy: 0.8811 - val_loss: 0.0894
Epoch 8/20
35/35 ————— 2s 51ms/step - accuracy: 0.9893 - loss: 0.0151 - val_accuracy: 0.8775 - val_loss: 0.0925
Epoch 9/20
35/35 ————— 2s 36ms/step - accuracy: 0.9911 - loss: 0.0124 - val_accuracy: 0.8759 - val_loss: 0.0951
Epoch 10/20
35/35 ————— 1s 36ms/step - accuracy: 0.9924 - loss: 0.0105 - val_accuracy: 0.8732 - val_loss: 0.0973
Epoch 11/20
35/35 ————— 1s 39ms/step - accuracy: 0.9935 - loss: 0.0090 - val_accuracy: 0.8741 - val_loss: 0.0989
Epoch 12/20
35/35 ————— 3s 49ms/step - accuracy: 0.9942 - loss: 0.0079 - val_accuracy: 0.8731 - val_loss: 0.1004
Epoch 13/20
35/35 ————— 2s 60ms/step - accuracy: 0.9946 - loss: 0.0071 - val_accuracy: 0.8715 - val_loss: 0.1020
Epoch 14/20
35/35 ————— 2s 36ms/step - accuracy: 0.9949 - loss: 0.0064 - val_accuracy: 0.8713 - val_loss: 0.1033
Epoch 15/20
35/35 ————— 1s 38ms/step - accuracy: 0.9953 - loss: 0.0059 - val_accuracy: 0.8715 - val_loss: 0.1043
Epoch 16/20
35/35 ————— 1s 34ms/step - accuracy: 0.9953 - loss: 0.0056 - val_accuracy: 0.8708 - val_loss: 0.1051
Epoch 17/20
35/35 ————— 1s 36ms/step - accuracy: 0.9954 - loss: 0.0053 - val_accuracy: 0.8699 - val_loss: 0.1058
Epoch 18/20
35/35 ————— 1s 32ms/step - accuracy: 0.9955 - loss: 0.0051 - val_accuracy: 0.8701 - val_loss: 0.1064
Epoch 19/20
35/35 ————— 1s 39ms/step - accuracy: 0.9955 - loss: 0.0050 - val_accuracy: 0.8700 - val_loss: 0.1069
Epoch 20/20
35/35 ————— 1s 37ms/step - accuracy: 0.9956 - loss: 0.0048 - val_accuracy: 0.8695 - val_loss: 0.1074

print(train_label[:10])

[1 0 0 1 0 0 1 0 1 0]

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